Supplement 1: Searches for VWD diagnosis guidelines

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We will combine these search terms to construct search strategies using **operators like AND and OR**.

We will use OR to combine the terms within each concept (e.g. Patients with VWD, bleeding assessment tools, VWF activity assays identified with a light grey shade in Table 1), to make sure we capture all the relevant articles to that specific concept, regardless of how the researchers labeled it. For example, when searching for articles about desmopressin trials, the search string will be "Deamino Arginine Vasopressin or desmopressin or DDAVP".

We will use AND to combine different concepts, to make sure to capture articles that are addressing the questions of interest. For example, for the question regarding bleeding assessment tools, we will combine the concepts "patients with VWD AND intervention bleeding assessment tools".

For some questions, the search will have both operators. For example, for the question regarding Patients with suspected Type 2A, 2B or 2M in need of additional testing, the strategy will combine patients with VWD AND (intervention VWF multimer testing) OR (intervention Collagen binding testing). Table 2 shows details of this.

We will also use other strategies for some concepts, such as **proximity of words**. For example, for patients with VWD, we will also search for abstracts that have the terms "Von Willebrand" close to the terms "disease OR disorder".

In addition, we will use **truncations and wild cards**- which are illustrated in the table. A truncation searches for all terms that are the same to the one specified up to the truncation. For example, searching for "bleed*" and truncating it will search for bleed, bleeding, bleeder, etc.

Finally, we will use **study design filters** to only capture randomized clinical trials and any type of observational studies comparing at least 2 of the options of interest, in addition to a filter to only capture articles related to diagnosis.

We will not limit the search by outcomes measured in the studies, date, or language of publication.

Table 1: Search terms used for each of the concepts

Term	It also includes
Patients with VWD	
Von Willebrand Disease	
Von Willebrand Disorder	
VWD	
Intervention: Bleeding assessment tools	
Bleeding assessment tools	
Bleeding score	
Screening tool	
questionnaire	
Interview	
Intervention: VWF activity assays	
Von Willebrand Factor	
VWF	
Ristocetin cofactor	VWF:RCo
VWF:G1b	VWF:G1bM VWF:GP1bR
Assay	
Glycoprotein 1b	
Glycoprotein Ib	
Intervention: Diagnosis exclusion	
Phenotype	
Evolving	
Aging	
Intervention: VWF propeptide trial	
VWF Propeptide	
VWFpp	
Intervention: Desmopressin trial	
Desmopressin trial	
Desmopressin challenge	
DDVAP	
Deamino Arginine Vasopressin	
Intervention: VWF multimer testing	
multimer	
Intervention: Collagen binding testing	
Collagen binding	
VWF:CB	
Intervention: Ristocetin induced platelet binding(RIPA) testing	
Ristocetin induced platelet binding	
RIPA	
PT-VWD	
Intervention: Genetic testing	
Genetic test	Genetic Assay Genetic Screen Genetic Assessment
Mutation	
Variant	
Genotype	
Gene	

Inheritance	
Intervention: VWF:FVIII binding	
VWF:FVIII binding	
VWF:FVIIIB binding	
Factor 8 binding	

Table 2: Concepts used for each question

Question	**Concepts**
BAT in patients suspected of VWD	patients with VWD AND Intervention Bleeding assessment tools
Patients suspected of VWD with negative BAT	patients with VWD AND intervention Bleeding assessment tools
Essays in patients with suspected VWD	patients with VWD AND intervention VWF activity assays
Patients with a historic diagnosis of VWD but who now have normal VWF levels	patients with VWD AND intervention Diagnosis exclusion
Patients with abnormal initial screen of VWF:Ag, VWF:RCo, and suspected Type 1 VWD cut-off values	patients with VWD AND intervention VWF activity assays
Propeptide levels and desmopressin trials in patients suspected of Type 1 VWD	patients with VWD AND (intervention VWF propeptide trial) OR (intervention Desmopressin trial)
Patients with abnormal initial screen of VWF:Ag, VWF:RCo and a low RCo:Ag ratio	patients with VWD AND intervention VWF activity assays
Patients with suspected Type 2A, 2B or 2M in need of additional testing	patients with VWD AND (intervention VWF multimer testing) OR (intervention Collagen binding testing)
Patients suspected to have Type 2B VWD	patients with VWD AND (intervention Ristocetin induced platelet binding) OR (intervention Genetic testing)
Patients suspected to have Type 2N in need of additional testing	patients with VWD AND (intervention VWF:FVIII binding) OR (intervention Genetic testing)

Example of search strategy: Bleeding assessment tools in patients suspected of VWD.

1. exp von Willebrand Diseases/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp.
2. (bleeding assess* tool* or bleed* score or screen* tool* or questionnaire* or interview).mp.
3. (bleed* adj4 (assess* or screen* or tool*)).mp.a
4. 2 or 3
5. 1 and 4

*The search strategy contains the following, which were not described above: “exp term /”, searches for all the terms indexed under the term (for example, exploding the term “Von Willebrand Diseases” searches also for VWD type 1, type 2, type 3 etc; “mp.”, searches for the term in multiples fields, including title, abstract, keywords, etc; “adj4”, is the tool to search for terms close to one another (in line 1, we searched for von willebrand term within 4 words of any of the 2 second terms (disease or disorder).

PICO 1: In Patients with suspected VWD - Using a bleeding assessment tool (any tool) to screen for VWD VS Non-standardized clinical assessment (not using a bleeding assessment tool)

Database(s): OVID Medline Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE(R) Daily and Ovid MEDLINE(R) 1946 to Present
Search Strategy:

#	Searches	Results
1	exp von Willebrand Diseases/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp.	6470
2	(bleeding assess* tool* or bleed* score or screen* tool* or questionnaire* or interview).mp.	784636
3	(bleed* adj4 (assess* or screen* or tool*)).mp.	2740
4	2 or 3	787072
5	1 and 4	314

Database(s): **Embase** 1974 to 2018 December 13
Search Strategy:

#	Searches	Results
1	exp von Willebrand Disease/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp.	10275
2	(bleeding assess* tool* or bleed* score or screen* tool* or questionnaire* or interview).mp.	1094952
3	(bleed* adj4 (assess* or screen* or tool*)).mp.	5047
4	2 or 3	1099275
5	1 and 4	853

PICO 2: Patients suspected of VWD with negative BAT (men and children) - Blood testing VS No blood testing

Database(s): OVID Medline Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE(R) Daily and Ovid MEDLINE(R) 1946 to Present
Search Strategy:

#	Searches	Results
1	exp von Willebrand Diseases/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp.	6470
2	(bleeding assess* tool* or bleed* score or screen* tool* or questionnaire* or interview).mp.	784636
3	(bleed* adj4 (assess* or screen* or tool*)).mp.	2740
4	2 or 3	787072
5	(blood adj4 test*).mp.	67402
6	4 or 5	850438
7	1 and 6	913

Database(s): **Embase** 1974 to 2018 December 13
Search Strategy:

#	Searches	Results
1	exp von Willebrand Disease/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp.	10275
2	(bleeding assess* tool* or bleed* score or screen* tool* or questionnaire* or interview).mp.	1094952
3	(bleed* adj4 (assess* or screen* or tool*)).mp.	5047
4	2 or 3	1099275
5	(blood adj4 test*).mp.	113728
6	4 or 5	1205027
7	1 and 6	1258

PICO 3: In Patients with suspected VWD - VWF:RCo (Automated and non-automated assays) VS Newer assays that reflect the platelet binding activity of VWF function (ie: VWF:G1bM, VWF:GP1bR).

Database(s): OVID Medline Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE(R) Daily and Ovid MEDLINE(R) 1946 to Present
Search Strategy:

#	Searches	Results
1	exp von Willebrand Diseases/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp.	6470
2	exp von willebrand factor/ or VWF.mp. or ristocetin cofactor.mp. or GPIb*.mp. or glycoprotein 1b.mp. or glycoprotein Ib.mp. or assay*.mp. or activity.mp.	3448167
3	1 and 2	3376

Database(s): **Embase** 1974 to 2018 December 13
Search Strategy:

#	Searches	Results
1	exp von Willebrand Disease/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp.	10275
2	exp von willebrand factor/ or (VWF or ristocetin cofactor or glycoprotein 1b or glycoprotein Ib or GPIb* or assay*).mp.	1602981
3	1 and 2	5758

PICO 4: Patients with a historic diagnosis of VWD but who now have normal VWF levels - Remove the diagnosis of VWD VS Reconsider another diagnosis

Database(s): OVID Medline Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE(R) Daily and Ovid MEDLINE(R) 1946 to Present
Search Strategy:

#	Searches	Results
1	exp von Willebrand Diseases/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp.	6470
2	(Phenotype or evolving or aging).mp.	804294
3	1 and 2	461

Database(s): **Embase** 1974 to 2018 December 13
Search Strategy:

#	Searches	Results
1	exp von Willebrand Disease/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp.	10275
2	(Phenotype or evolving or aging).mp.	1189124
3	1 and 2	1034

Notes: Subgroup: Bleeding phenotype

PICO 5: Patients with abnormal initial screen of VWF:Ag, VWF:RCo, and suspected Type 1 VWD - Diagnosis of Type 1 VWD at VWF:Ag and/or VWF:RCo < 0.30 IU/mL VS Diagnosis of Type 1 VWD at VWF :Ag and/or VWF:RCo < the reference range of your lab which is usually 0.50 IU/mL

Database(s): OVID Medline Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE(R) Daily and Ovid MEDLINE(R) 1946 to Present
Search Strategy:

#	Searches	Results
1	exp von Willebrand Diseases/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp.	6470
2	exp von willebrand factor/ or VWF.mp. or ristocetin cofactor.mp. or GPIb*.mp. or glycoprotein 1b.mp. or glycoprotein Ib.mp. or assay*.mp. or activity.mp.	3448167
3	1 and 2	3376

Database(s): **Embase** 1974 to 2018 December 13
Search Strategy:

#	Searches	Results
1	exp von Willebrand Disease/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp.	10275
2	exp von willebrand factor/ or (VWF or ristocetin cofactor or glycoprotein 1b or glycoprotein Ib or GPIb* or assay*).mp.	1602981
3	1 and 2	5758

Notes: VWF:RCo means activity essays in general (ie: VWF:G1bM, VWF:GP1bR)

PICO 6: Patients suspected of Type 1 VWD – VWFpp VS Desmopressin trial with 1 and 4 hour bloodwork

Database(s): OVID Medline Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE(R) Daily and Ovid MEDLINE(R) 1946 to Present
Search Strategy:

#	Searches	Results
1	exp von Willebrand Diseases/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp.	6470
2	(propeptide or VWFpp).mp.	5897
3	Deamino Arginine Vasopressin/ or desmopressin.mp. or DDAVP.mp.	5690
4	(challenge or trial).mp.	1383647
5	3 and 4	736
6	2 or 5	6625
7	1 and 6	182

Database(s): **Embase** 1974 to 2018 December 13
Search Strategy:

#	Searches	Results
1	exp von Willebrand Disease/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp.	10275
2	(propeptide or VWFpp).mp.	7864
3	"argipressin[1 deamino]"/ or desmopressin.mp. or DDAVP.mp.	13200
4	(challenge or trial).mp.	2282560
5	3 and 4	2119
6	2 or 5	9976
7	1 and 6	535

PICO 7: Patients with abnormal initial screen of VWF:Ag, VWF:RCo and a low RCo:Ag ratio - RCo:Ag <0.5 VS Other cut-offs (ie: 0.6 or 0.7)

Database(s): OVID Medline Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE(R) Daily and Ovid MEDLINE(R) 1946 to Present
Search Strategy:

#	Searches	Results
1	exp von Willebrand Diseases/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp.	6470
2	exp von willebrand factor/ or VWF.mp. or ristocetin cofactor.mp. or GPIb*.mp. or glycoprotein 1b.mp. or glycoprotein Ib.mp. or assay*.mp. or activity.mp.	3448167
3	1 and 2	3376

Database(s): **Embase** 1974 to 2018 December 13
Search Strategy:

#	Searches	Results
1	exp von Willebrand Disease/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp.	10275
2	exp von willebrand factor/ or (VWF or ristocetin cofactor or glycoprotein 1b or glycoprotein Ib or GPIb* or assay*).mp.	1602981
3	1 and 2	5758

Notes: VWF:RCo means activity essays in general (ie: VWF:G1bM, VWF:GP1bR)

PICO 8: Patients with suspected Type 2A, 2B or 2M in need of additional testing - VWF multimer analysis VS VWF:CB

Database(s): OVID Medline Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE(R) Daily and Ovid MEDLINE(R) 1946 to Present
Search Strategy:

#	Searches	Results
1	exp von Willebrand Diseases/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp.	6470
2	(multimer or Collagen binding or VWF:CB).mp.	3926
3	1 and 2	477

Database(s): **Embase** 1974 to 2018 December 13
Search Strategy:

#	Searches	Results
1	exp von Willebrand Disease/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp.	10275
2	(multimer or Collagen binding or VWF:CB).mp.	5457
3	1 and 2	941

PICO 9: Patients suspected to have Type 2B VWD - Ristocetin induced platelet binding(RIPA) VS Genetic testing (mutation analysis)

Database(s): OVID Medline Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE(R) Daily and Ovid MEDLINE(R) 1946 to Present
Search Strategy:

#	Searches	Results
1	exp von Willebrand Diseases/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp.	6470
2	(ripa or ristocetin induced platelet binding or PT-VWD).mp.	501
3	((genetic or mutation or variant or genotype or gene or inheritance) adj4 (test* or assay* or screen* or assess*)).mp.	163485
4	2 or 3	163976
5	1 and 4	275

Database(s): **Embase** 1974 to 2018 December 13
Search Strategy:

#	Searches	Results
1	exp von Willebrand Disease/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp.	10275
2	(ripa or ristocetin induced platelet binding or PT-VWD).mp.	873
3	((genetic or mutation or variant or genotype or gene or inheritance) adj4 (test* or assay* or screen* or assess*)).mp.	206516
4	2 or 3	207360
5	1 and 4	517

PICO 10: Patients suspected to have Type 2N in need of additional testing - VWF:FVIII binding VS Genetic testing.

Database(s): OVID Medline Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE(R) Daily and Ovid MEDLINE(R) 1946 to Present
Search Strategy:

#	Searches	Results
1	exp von Willebrand Diseases/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp.	6470
2	((factor VIII or VWF:FVIII or factor 8) adj4 binding).mp.	323
3	((genetic or mutation or variant or genotype or gene or inheritance) adj4 (test* or assay* or screen* or assess*)).mp.	163485
4	2 or 3	163802
5	1 and 4	253

Database(s): **Embase** 1974 to 2018 December 13
Search Strategy:

#	Searches	Results
1	exp von Willebrand Disease/ or VWD.mp. or (von willebrand adj4 (disease or disorder)).mp.	10275
2	((factor VIII or VWF:FVIII or factor 8) adj4 binding).mp.	426
3	((genetic or mutation or variant or genotype or gene or inheritance) adj4 (test* or assay* or screen* or assess*)).mp.	206516
4	2 or 3	206925
5	1 and 4	436

Study Design Filters

Database(s): OVID Medline Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE(R) Daily and Ovid MEDLINE(R) 1946 to Present
Search Strategy:

#	Searches	Results
29	limit 28 to human	3276
30	(randomized controlled trial or controlled clinical trial).pt. or (randomized or placebo).ab. or drug therapy.fs. or (randomly or trial or groups).ab.	4337050
31	epidemiologic studies/ or exp case control studies/ or exp cohort studies/ or Cross-sectional studies/	2225583
32	case control.tw.	111970
33	(cohort adj (study or studies)).tw.	166398
34	cohort analy$.tw.	6624
35	(Follow up adj (study or studies)).tw.	46061
36	(Observational adj (study or studies)).tw.	87158
37	(longitudinal or retrospective or cross sectional).tw.	921384
38	or/31-36	2366260
39	(registr* or database).mp.	468967
40	(sensitiv: or diagnos:).mp. or di.fs.	5730164
41	30 or 38 or 39 or 40	10134762
42	29 and 41	1939

Database(s): **Embase** 1974 to 2018 December 13
Search Strategy:

#	Searches	Results
29	limit 28 to human	5475
30	di.fs. or predict:.tw. or specificity.tw.	4873662
31	random*.ab. or random*.ti. or (clinical adj trial*).hw,ab,ti. or exp 'health care quality'/	4680717
32	exp disease course/ or risk:.mp. or diagnos:.mp. or follow-up.mp. or ep.fs. or outcome.tw.	11189515
33	30 or 31 or 32	14073954
34	29 and 33	3781

Supplement 2: Excluded studies

In patients suspected of VWD, should we use VWF:RCo (Automated and non-automated assays) or use newer assays that reflect the platelet binding activity of VWF function (ie: VWF:G1bM, VWF:GP1bR)?

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In patients suspected of Type 1 VWD with increased VWF clearance (e.g. Type 1C), should we use ratio of VWF propeptide to VWF antigen (VWFpp/VWF:Ag) or use desmopressin trial with 1 and 4 hour bloodwork?

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In patients suspected of Type 2A, 2B or 2M VWD in need of additional testing, should we use VWF multimer analysis or use VWF collagen binding (VWF:CB) to VWF antigen ratio (VWF:CB/VWF:Ag)?

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In patients suspected of Type 2B VWD, should we use Ristocetin induced platelet aggregation/agglutination (RIPA), or use genetic testing (mutation analysis)

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In patients suspected of Type 2N in need of additional testing, should we use VWF Factor VIII binding (VWF:FVIII binding), or use genetic testing (mutation analysis)

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Supplement 3: Risk of bias

Author	Risk of bias population selection	Risk of bias index test	Risk of bias reference test[a]	Flow and timing Risk of bias
Vangenechten, 2018	High	Low	Low	Low
Boender, 2018	Moderate	Low	High	Low
Sagheer, 2016	High	Low	Low	Low
Costa Pento, 2014	High	Low	Low	Low
Verfaillie, 2013	High	Low	Low	Low
Cabrera, 2013	High	Low	Moderate	Low
Lasne, 2012	High	Low	Low	Low
Trossaert, 2011	Low	Low	Low	Low
Chen, 2011	High	Low	Low	Low
Salem, 2007	Low	Low	Low	Low
Pinol, 2007	High	Low	Low	Low
Vleeschauwer, 2006	High	Low	Low	Low
Strandberg, 2006	High	Low	Low	Low

Author	Risk of bias population selection	Risk of bias index test	Risk of bias reference test[a]	Flow and timing Risk of bias
Sztukowska, 2008	High	Low	Low	Low
Haberichter, 2006	Low	Low	High	Low
Eikenboom, 2013	Low	Low	High	Low
Haberrichter, 2008	Low	Low	High	Low
Stufano, 2019	Low	Low	Low	Low

a. Reference test determining patients with increased clearance not clearly defined.

Author	Risk of bias population selection	Risk of bias index test	Risk of bias reference test[a]	Flow and timing Risk of bias
Rodriguez, 2018	Moderate	Low	Low	Low
Vangenechten, 2018	High	Low	Low	Low
Jousselme, 2018	High	Low	Low	Low
Bowyer, 2018	High	Low	Low	Low
Casonato, 2017	Low	Moderate	Low	Low

Ni, 2013	Low	Low	Low	Low
Flood, 2013	High	Low	Moderate	Low
Popov, 2006	Low	Low	Low	Low
Adcock, 2006	Low	Low	Low	Low
Riddell, 2002	High	Low	High	Low
Federici, 2000	High	Moderate	Low	Low

Author	Risk of bias population selection	Risk of bias index test	Risk of bias reference test	Flow and timing Rsk of bias
Woods, 2017	High	Low	Low	Low
Borras, 2017	Low	Low	High	Low
Veyradier, 2016	High	Low	High	Low
Shen, 2016	Low	High	Low	Low
Battle, 2016	Low	Low	High	Low
Laderas, 2015	Low	High	High	Low
Kaur, 2014	High	High	High	Low
Hamilton, 2011	High	High	Low	Low
Federici, 2009	High	Low	Low	Low
Caron, 2006	High	Low	Low	Low
Facey, 2000	High	Low	Low	Low
Casana, 1998	High	High	Low	Low
Wood, 1996	Low	Low	High	Low
Cooney, 1991	High	Low	High	Low

Author	Risk of bias population selection	Risk of bias index test	Risk of bias reference test	Flow and timing Rsk of bias
Battle, 2006	High	Low	High	Low
Borras, 2017	Low	High	Low	Low
Casonato, 2018	Low	Low	High	Low
Veyradier, 2016	High	High	Low	Low
Costa Pinto, 2014	High	High	Low	Low
Wang, 2013	Low	High	Low	Low
Hamoshire, 2013	High	High	Low	Low
Veyradier, 2011	High	Low	High	Low
Zhukov, 2009	High	Low	High	Low
Coralles, 2009	High	High	High	Low
Casanato, 2007	High	Low	High	Low

Taylor, 2002	Low	Low	High	Low
Rodgers, 2002	High	High	Low	Low
Caron, 2002	High	Low	Low	Low
Casonato, 1998	High	Low	High	Low
Bowen, 1998	Low	High	High	Low
Schneppenheim, 1996	Low	Low	High	Low

Supplement 4: Supplement data

Table 1: Identified mutations in type 2B VWD

Studies	Mutations Identified
Freitas, 2019	R1341Q, R1308C, P1266L
Woods, 2017	Y1258C, P1266L,M1304V, R1306W, R1308C, S1310F, V1316M
Borras, 2017	R1306W, R1308R, V1316M, P1266L, P1266Q, R1306Q
Veyradier, 2016	M1304dup, R1306Q/P/W, R1308C, I1309V, S1310F/P, V1316M, P1337L, R1341L/Q/W, I1372S, L1360F/P, A1461V, P1266L/Q, H1268D/N/Q
Shen, 2016	R1306W, V1316M, R1308C
Batlle, 2016	R1308C, R1306W, V1316M, P1266L, R1306Q, Pro1266Q
Laderas, 2015	R1306Q, R1306W, R1308C, R1315H, R1341Q
Kaur, 2014	R1341Q, H1268N, V1316M, R1306W
Federici, 2009	P1266Q/L, H1268D, R1306W, R1308C/L, I1309V, V1316M, P1337L, R1341Q/W
Caron, 2006	H1268D, R1306W, R1306Q, R1306L, R1308C, V1316M, R1341Q, A1461V
Facey, 2000	R543W, R545C, R543L
Casana, 1998	R1308C, V1316M, P1337L, R1306W, R1341W
Wood, 1996	R543Q, Val553M, S547F, R578Q
Cooney, 1991	R543Q, R545C, Val553M, and R578Q

Table 2: Phenotype-genotype correlation results for type 2B VWD

Studies	Correlation
Woods, 2017	p.P1266L, 100% at 0.4 mg/ml p.M1304V, 66.6% at 0.5 and 33.4% at 0.4 p.R1306W, 20% at 0.6, 0.5, 0.4 and 40% at 0.3 p.R1308C, 25% at 0.7, 50% at 0.6, 12.5% at 0.4, 12.5% at 0.2 p.S1310F, 33.3% at 0.7 and 66.6% at 0.4 p.V1316M, 10% at 0.7, 0.6, 0.3, and 50% at 0.5, and 20% at 0.2 p.Y1258C, 100% at 0.3
Borras, 2017	- 35 patients were diagnosed as having type 2B VWD by molecular diagnosis. - A good phenotype-genotype correlation could be established **for all patients**, as most showed a loss of high-molecular-weight multimers and discordance between VWF:Ag and VWF:RCo levels (mean ratio=0.51; range 0.19-1.1) and a classical type 2B mutation.
Veyradier, 2016	100% correlation for 112 (17%) that exhibited type 2B VWD including 95 patients with a ‘‘classical’’ type 2B and 17 patients with a type 2B ‘‘New York’’
Batlle, 2016	- 35 patients were diagnosed as having type 2B VWD by molecular diagnosis. - A good phenotype-genotype correlation could be established for all patients, as most showed a loss of high-molecular-weight multimers and discordance between VWF:Ag and VWF:RCo levels (mean ratio=0.51; range 0.19-1.1) and a classical type 2B mutation.
Laderas, 2015	RIPA was positive in 3 of 5 mutations identified p.R1306Q, p.R1306W, and p.R1308C
Hamilton, 2011	48/110 had A1 domain mutations consistent with type 2B VWD. Seventeen cases carried platelet *GP1BA* mutations consistent with PT-VWD. In both the Australian and UK cases, apart from normal family members, there has not been any case where the phenotypic diagnosis has not matched the genotype finding, and ultimately making either a correct type 2B VWD, or its alternative PT-VWD, diagnosis possible. In Brazil, RIPA was performed in 14/18 cases, showing an enhanced response in 12, yet genetic analysis identified 2B VWD mutations in only three cases. In Canada, Apart from two normal family members, 9/40 were mutation negative for both *VWF and GP1BA* and only two cases showed GP1BA mutations. In the three cases from Switzerland, the 2B VWD phenotype matched the genetic analysis identifying known 2B VWD mutations. In Sweden, one case had a VWF mutation R1308P coinciding with 2B VWD and enhanced RIPA, but both parents of this index cases were completely normal, both phenotypically and genotypically with respect to the VWF gene.
Federici, 2009	All mutations were captured at a mean RIPA concentration of 0.6 mg/mL (0.3-0.8)
Caron, 2006	All 31 cases displayed a positive RIPA at 0.5 mg/ml ristocetin concentration.
Facey, 2000	In all cases, RIPA occurred at concentrations of 0.5 mg/ml of ristocetin, while in one individual it occurred at 0.25 mg/ml of ristocetin. The RIPA results demonstrated increased platelet sensitivity to reduced levels of ristocetin, a finding consistent with type 2B VWD.
Wood, 1996	RIPA was increased in all 7 patients studied, but the concentration is not indicated in the study.
Cooney, 1991	All mutations were captured as patients had enhanced RIPA at a low concentration of ristocetin (0.2-0.6mg/ml)

Table 3: Genetic variant detection for type 2N VWD

Article	Mutations
Borras, 2017	R816W, R854Q
Batlle, 2016	R816W and R854Q
Casonato, 2018	R854Q, P812Rfs*31, C524Y, R760C
Veyradier, 2016	R768Q, C788Y, T791M, L809P, R816W, R854Q, G887R, C1060R
Costa Pinto, 2014	R816W, R854Q
Wang, 2013	P812L, R854Q, R924Q
Hamoshire, 2013	C788R, C1225G
Veyradier, 2011	R816W, R854Q, C788Y, C1070R, T791M, L884FfsX19, C1060R
Zhukov, 2009	R854Q, H817Q, H817Q/R1342C
Corrales, 2009	R816W
Casanato, 2007	R854Q, R760C
Taylor, 2002	R53W, R91Q
Rodgers, 2002	R854Q
Caron, 2002	R816W, R854Q, C858F, C804F,
Casonato, 1998	R53W, R91Q
Bowen, 1998	R854Q, R952Q, R816W, H817Q, C858F
Schneppenheim, 1996	E24K, T28M, R91Q

Table 4: Phenotype-genotype correlation results for type 2N VWD

Author, year	Results: In all studies homozygous VWD2N patients had binding ratios <0.12, heterozygous carriers had intermediate binding ratios of 0.44–0.61, and healthy control subjects had ratios of 0.73–1.42
Batlle, 2016 Borras, 2017	9 patients were type 2N VWD and 11 patients were carriers of type 2N VWD. Regarding the patients with 2N VWD, five were homozygous for a 2N mutation, and four were compound heterozygous in trans for p.Gln895His and a nonsense or missense mutation.
Casonato, 2018	Genetic analysis demonstrated that all the patients with VWF:FVIIIB ratios below 0.3 were carrying the p.R854Q mutation at homozygous or compound heterozygous level with a quantitative VWF defect. There were also 51 patients with a VWF:FVIIIB ratio below 0.74, but above 0.3. 34/51 (67%) were heterozygous for the p.R854Q mutation, and one was carrying the p.R760C mutation at heterozygous level; two of the 34 patients were also haemophilia A carriers, and one suffered from haemophilia A. The other 16 patients revealed no mutations in the main FVIII binding domain of VWF.
Veyradier, 2016	100% correlation for the 81 type 2N VWD patients. in type 2N, 22 truncating mutations leading to a silent allele were also found (type 2N/3 patients) including one-third also found in our type 3 VWD patients
Costa Pinto, 2014	All type 2N VWD patients (n = 5) showed normal VWF:RCo/VWF:Ag ratios and VWF:FVIIIB <0.8.
Hamoshire, 2013	When compared against normal plasma (100%), patient plasma had reduced FVIII binding capacity (VWF:FVIIIB) similar to that observed in plasma from a known type 2N patient homozygous for p.T791M (p.C788R homozygote 0% (heterozygote 49%) vs. 1.3%; p.C1225G homozygote 7.1% vs. 7.1%)
Veyradier, 2011	The mean FVIII:C/VWF:Ag ratio is 0.26 which is mainly representative of the homozygous Arg854Gln subgroup (mean FVIII:C/ VWF:Ag ratio at 0.27). The values range between 0.04 and 0.47. 9 heterozygous carriers for a 2N mutation includes 8 subjects with a Arg854Gln mutation and 1 subject with a Cys1060Arg mutation. In all of them, the FVIII:C/VWF:Ag ratio is normal, higher than 0.6. All patients with type 2N VWD exhibit a severely decreased VWF:FVIIIB with values lower than 15%, and No control subject (healthy subjects, haemophilia A, haemophilia carriers or VWD patients other than type 2N) exhibit a markedly decreased VWF:FVIIIB.
Zhukov, 2009	All samples from subjects with homozygous or heterozygous mutation showed abnormal VWF- FVIII binding, and three distinct ratio ranges were observed: homozygous VWD2N patients had binding ratios <0.12, heterozygous carriers had intermediate binding ratios of 0.44–0.61, and healthy control subjects had ratios of 0.73–1.42 Special precautions must be taken when reporting patient results in the 0.65–0.72 range, which is probably the assay's true equivocal zone; rare outliers of both normal and heterozygous individuals occasionally fall in this range, as do results from compromised samples.
Casanato, 2007	All the type 2N carriers identified in the present study had a reduced VWF:FVIIIB to VWF:Ag ratio, regardless of the FVIII/VWF:Ag ratio or VWF:FVIIIB values. The mean VWF:FVIIIB ratio was 0.56±0.10 vs normal >0.75 and no relationship was demonstrable between VWF:FVIIIB and FVIII/VWF:Ag.
Taylor, 2002	The homozygous R53W sample exhibited minimal FVIII binding activity, whilst the heterozygous R91Qr gave a result of 0.43 compared with the PNP reference plasma value of 1.0
Rodgers, 2002	Patients with very low factor VIII binding were clearly identified, and all control subjects with hemophilia were clearly identified as having normal factor VIII binding.
Caron, 2002	A total of 15 unrelated patients were diagnosed as being affected with type 2N VWD because their VWF:FVIIIB was found to be markedly decreased (9.65 ± 2.75%, n = 14). 5 patients exhibited intermediate FVIII binding capacity (VWF:FVIIIB = 57.2 ± 6.8%), similar but slightly greater (P = 0.015) than that obtained with the NP/2N mixture.
Schneppenheim, 1996	All 5 patients and their families (total of 68) with VWD type 2N homogenous and heterogenous mutations had a VWF:FVIII level of <60, (if homogenous <8) except for 1 patient with WT|R91Q genotype had a level of 63.

Table 5: VWF activity level assays used and reported correlations between assays

Author	Year	Pts	Assay 1	Assay 2	Assay 3	Cor 1-2	Cor 2-3	Cor 1-3
Boender, J	2018	618	BC-VWF:Rco	VWF:GPIbR	VWF:GPIbM	0.957	0.984	0.959
Szederjesi A	2018	95	BC-VWF:Rco	VWF:GPIbR	VWF:GPIbM	0.963		0.989
Sagheer, S	2016	60	VWF:RCo[Agg]	VWF:GPIbR	VWF:Ab	0.954		0.938
Favaloro E	2016	535	VWF:RCo[Agg]	VWF:GPIbR	VWF:GPIbM	0.928		0.942
Timm A	2015	170	BC-VWF:Rco	VWF:GPIbR	VWF:GPIbM	0.927	0.921	0.912
Stitt C	2014	37	BC-VWF:Rco	VWF:GPIbM		0.989		
Patzke, J	2014	580	BC-VWF:Rco	VWF:GPIbM		0.99		
Geisen, U	2014	432	BC-VWF:Rco	VWF:GPIbM		0.96		
Favaloro, E	2014	600	BC-VWF:Rco	VWF:GPIbM		0.958		
De Maistre	2014	122	VWF:RCo[Agg]	VWF:RCo[Acu]	VWF:GPIbM	0.977		0.965
Costa Pinto	2014	176	VWF:RCo[Agg]	VWF:RCo[Acu]		0.92		
Verfaillie, C	2013	50	VWF:GPIbR	VWF:Ab	VWF:RCo[Agg]	0.94		0.77
Lawrie, A	2013	180	BC-VWF:RCo	VWF:GPIbM		0.97		
Cabrera, N	2013	91	VWF:RCo[Agg]	VWF:GPIbR		0.92		
Trossaert, M	2011	268	VWF:Ab	BC-VWF:Rco		0.89		
Chen, D	2011	468	BC-VWF:Rco	VWF:Ab		0.93		
Bowyer, A	2011	53	VWF:RCo[Agg]	BC-VWF:Rco		0.91		
Chen, D	2008	35	VWF:Rco [Agg]	VWF:Rco (Flow Cyt)		0.86		
Pinol, M	2007	127	VWF:Rco [Agg]	VWF:Ab		0.956		
Vleeschauwer, A	2006	148	VWF:Rco [Agg]	VWF:Ab		0.84		
Sucker, C	2006	300	VWF:Ab	BC-VWF:Rco		0.88		
Strandberg, K	2006	478	VWF:Rco [Agg]	BC-VWF:Rco		0.96		
Vanhoorelbeke, K	2005	92	VWF:Rco [Agg]	VWF:RCo ELISA		0.963		
Lattuada, A	2004	95	VWF:Rco [Agg]	BC-VWF:Rco		0.61		
Federici, A	2004	122	VWF:Rco [Agg]	VWF:RCo ELISA		0.93		